AI Video Caption Generator

DOCUMENTATION — v1.0.0

Nekkanti Satya Srinath

1. Documentation Overview

This document provides consolidated technical and user-facing documentation for AI Video Caption Generator v1.0.0. The application generates multilingual captions from videos, creates SRT and VTT subtitle files, preserves the original video, and produces a separate captioned video using FFmpeg.

2. Project Purpose

The project is a modular, local-first video captioning application. Whisper provides timestamped transcription, language detection identifies the spoken language, Ollama with TranslateGemma performs translation, and FFmpeg burns captions into a separate output video.

3. Complete Workflow

Upload Video → Save Original → Whisper Transcription → Language Detection → Caption Language Selection → Translation → SRT/VTT → FFmpeg Burn → Captioned Video

4. Main Features

- Video upload and original-video preservation.
- Timestamped internal transcription with Whisper.
- Spoken-language detection.
- Multilingual caption generation.
- Local Ollama / TranslateGemma translation.
- SRT subtitle generation.
- VTT subtitle generation.
- Permanent caption burning with FFmpeg.
- Separate captioned-video output.
- Dashboard with statistics and recent files.
- Settings page for Whisper, Ollama and default caption language.
- Help and troubleshooting page.
- About page.
- Sidebar navigation.
- Automated PyTest test suite and GitHub Actions CI.

5. Technology Stack

- Python 3.11
- Streamlit
- OpenAI Whisper

- Ollama
- TranslateGemma 12B
- FFmpeg
- PyTest
- GitHub Actions

6. Architecture

The application separates UI, agents, services, providers, core models, configuration and utility functionality.

- UI / Streamlit — application presentation and interaction.
- Agents — workflow coordination and application-level behavior.
- Services — reusable processing logic.
- Providers — replaceable AI integrations.
- Core — caption data models and domain structures.
- Config — supported formats, languages and defaults.
- Utils — caption, subtitle and formatting helpers.

7. Core Workflow Components

- CaptionAgent — coordinates caption workflow operations.
- CaptionGenerationService — translates timestamped caption segments.
- LanguageDetectionService — detects spoken language.
- TranscriptService — produces timestamped internal transcript data.
- TranslationProvider — abstraction for translation providers.
- OllamaTranslationProvider — local Ollama translation implementation.
- Video caption burn service — uses FFmpeg to create captioned video output.
- DashboardAgent — provides application statistics and recent-file information.

8. Supported Formats

- Video: MP4, MOV, AVI, MKV and WebM.
- Caption: SRT and VTT.

9. Supported Caption Languages

- English (en)
- Telugu (te)
- Hindi (hi)
- Tamil (ta)
- Kannada (kn)
- Malayalam (ml)
- Bengali (bn)
- Marathi (mr)
- Gujarati (gu)

- Punjabi (pa)

10. Storage

- uploads/ — original uploaded videos.
- captions/ — generated SRT and VTT files.
- outputs/ — final captioned videos.
- Original videos are preserved rather than overwritten.
- Dashboard recent-file listings exclude .gitkeep placeholders.

11. UI Pages

- Dashboard
- Caption Generator
- Captions
- Settings
- Help
- About
- Sidebar navigation

12. Settings

- Whisper model selection.
- Ollama translation model configuration.
- Default caption language selection.
- Session-based application settings.

13. Translation

Caption translation is abstracted through TranslationProvider. The Ollama implementation sends caption text to a local Ollama API and requests only the translated text. The configured default model is TranslateGemma 12B.

14. Error Handling

- Empty caption language is rejected.
- Empty caption segment collections are rejected.
- Missing transcript service is reported.
- Missing translation configuration is reported.
- Ollama connection failures are reported.
- Empty translation responses are rejected.

15. Testing

The final automated suite was executed from Windows PowerShell using Python 3.11.4 and PyTest 9.1.1.

Metric	Result

Total tests	153
Passed	153
Failed	0
Pass rate	100%
Execution time	15.31 seconds
Final status	TESTED SUCCESSFULLY

16. Final Verification

- Original video saved successfully.
- SRT generated successfully.
- VTT generated successfully.
- Captioned video generated successfully.
- Dashboard verified.
- Settings verified.
- Help verified.
- About verified.
- Sidebar navigation verified.
- Requirements validation verified.

17. Documentation Set

- README.md
- API_DOCUMENTATION.md
- ARCHITECTURE.md
- CHANGELOG.md
- CODE_OF_CONDUCT.md
- CONFIGURATION.md
- CONTRIBUTING.md
- DOCUMENTATION.md
- EXPORT_GUIDE.md
- FAQ.md
- FEATURES.md
- IMPLEMENTATION_STEPS.md
- INSTALLATION.md
- INTERVIEW_ANSWERS.md
- INTERVIEW_QUESTIONS.md
- PROJECT_NOTES.md
- PROJECT_PLANNER.md
- PROJECT_STRUCTURE.md

- PROVIDER_GUIDE.md
- RELEASE_NOTES.md
- RESUME_BULLETS.md
- SECURITY.md
- SYSTEM_DESIGN.md
- TESTING.md
- TROUBLESHOOTING.md
- USER_GUIDE.md
- WORKFLOW.md
- SCREENSHOTS.md

18. GitHub and CI

GitHub Actions is used to execute the automated test suite. Project documentation includes GitHub Actions, test, status, version and license badges. The verified test baseline is 153 passing tests.

19. Release Status

Version: v1.0.0 — Core Release Complete

- 153 automated tests passing.
- 0 automated test failures.
- 100% pass rate.
- End-to-end workflow verified.
- Original, SRT, VTT and captioned-video outputs verified.

20. Future Enhancements

- Caption editor.
- Batch processing.
- Caption styling controls.
- Additional translation providers.
- Additional subtitle formats.
- Advanced FFmpeg controls.

21. Author

Nekkanti Satya Srinath

22. License

This project is licensed under the **MIT License**.